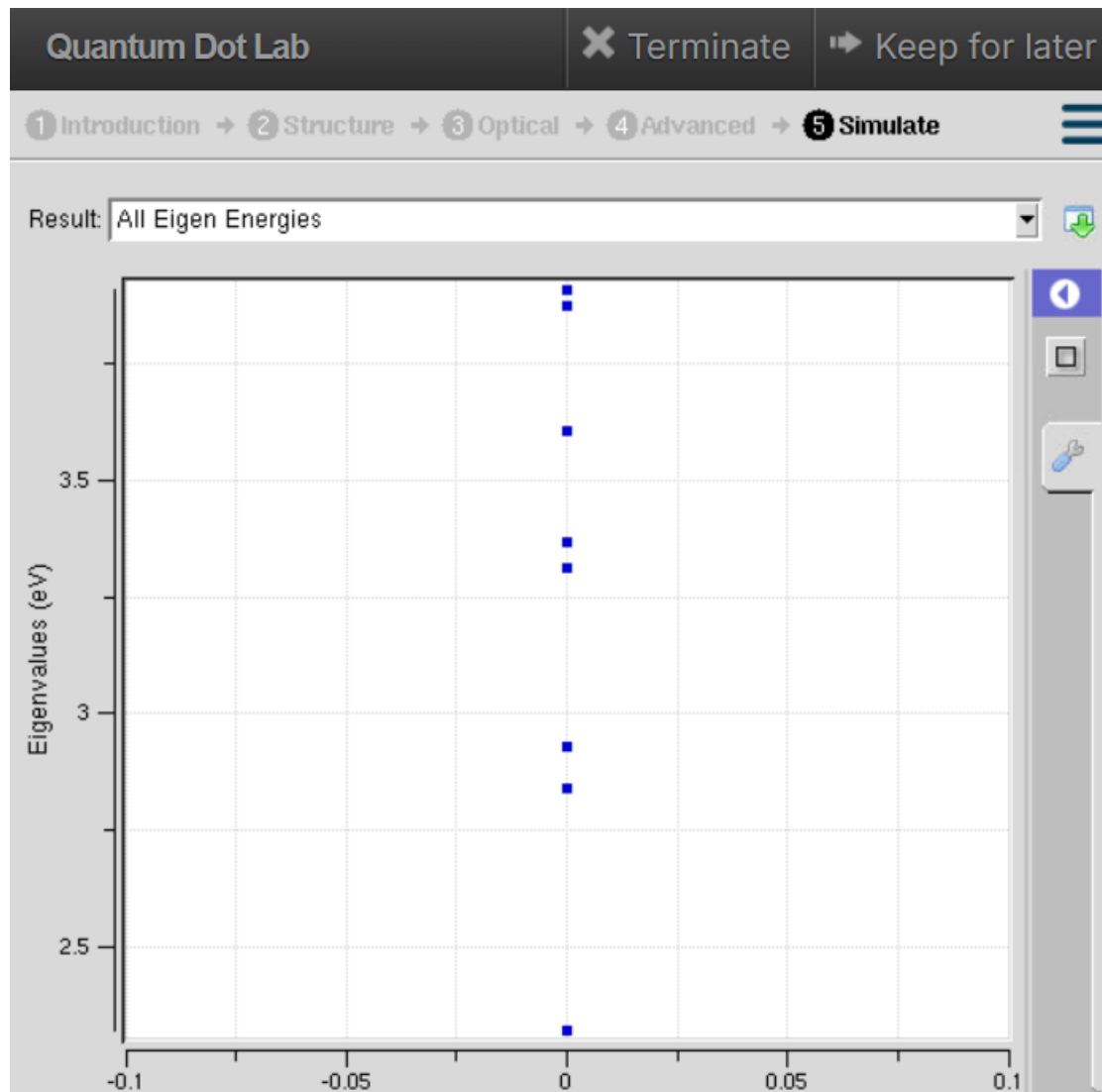
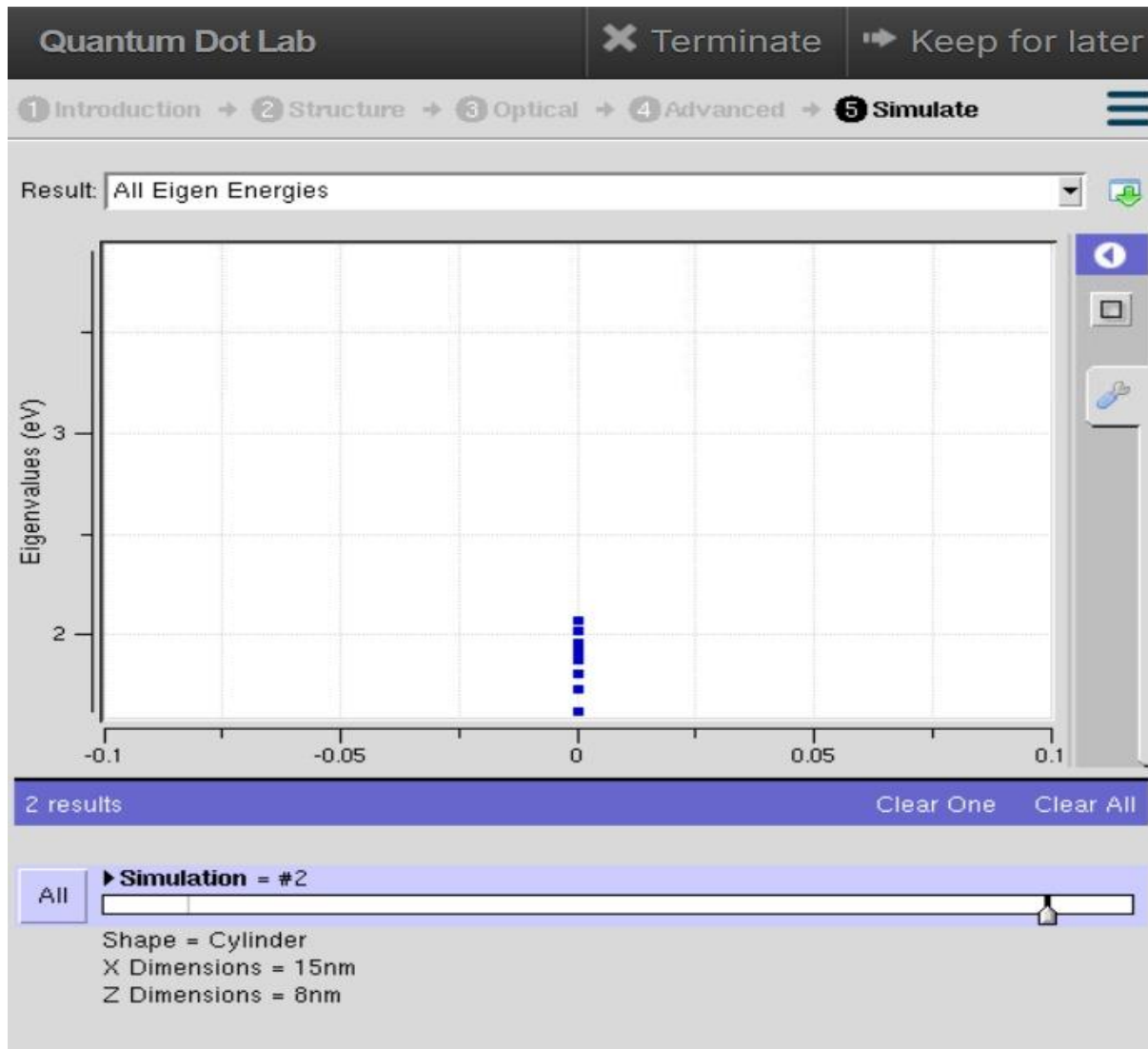


Exp-4 Simulation of Energy States of Quantum Dot using NanoHub



This image presents the **All Eigen Energies** obtained from the **Quantum Dot Lab** simulation. The graph displays the discrete **eigenvalue energy levels (in eV)** of electrons confined within the quantum dot, demonstrating the quantum mechanical behavior of the nanostructure. Unlike bulk semiconductors, quantum dots exhibit **discrete energy states** due to strong three-dimensional carrier confinement. Each blue point represents an allowed energy level that electrons can occupy, with values ranging approximately from **2.3 eV to 3.9 eV**. The spacing between these energy levels provides insight into the quantum confinement strength and electronic properties of the device. This analysis is essential for evaluating the optical and electrical performance of quantum dots and is widely used in the design of **quantum dot lasers, LEDs, photodetectors, solar cells, and quantum computing devices**, where precise control of energy levels is critical for achieving high efficiency and improved device performance.

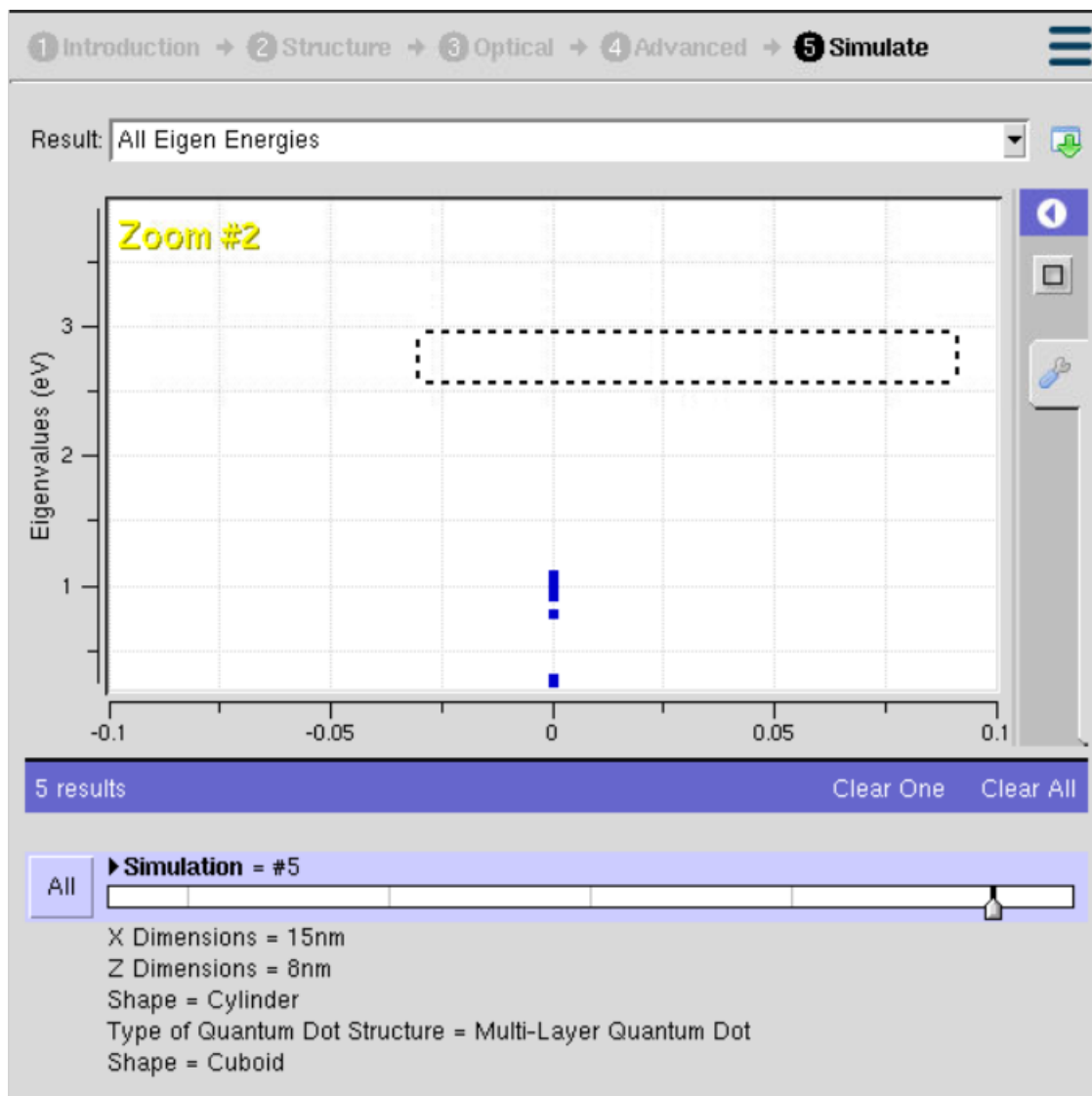
Case study-1: Simulate the energy states of cylindrical quantum dots of $X=15\text{ nm}$, $Y = 10.5\text{ nm}$ and $Z = 8\text{ nm}$.



This image shows the **second Quantum Dot Lab simulation** displaying the **All Eigen Energies** of a **cylindrical quantum dot** with dimensions of **15 nm (X-axis)** and **8 nm (Z-axis)**. The graph presents the discrete energy levels (eigenvalues) available to electrons inside the quantum dot, which are concentrated between approximately **1.6 eV and 2.1 eV**. Compared to the previous simulation, the energy levels are lower and more closely spaced, indicating that increasing the quantum dot dimensions reduces quantum confinement and allows electrons to occupy lower energy states. The simulation summary at the bottom confirms the cylindrical geometry and the modified dimensions used in **Simulation #2**. These results are important for optimizing the optical and electronic properties of quantum dots in applications such as **quantum dot lasers, light-emitting diodes (QLEDs), photodetectors, solar cells, and**

quantum computing, where controlling electron energy levels directly influences device efficiency and performance.

Case study-2: Simulate the energy states of two layer cuboid quantum dots



This image presents the **fifth Quantum Dot Lab simulation**, illustrating the **All Eigen Energies** of a **multi-layer quantum dot structure** with **15 nm (X-dimension)** and **8 nm (Z-dimension)**. The simulation combines a **cylindrical quantum dot** within a **cuboid multilayer structure**, allowing the analysis of electron confinement in a more complex nanoscale geometry. The plotted eigenvalues are concentrated between approximately **0.3 eV and 1.1 eV**, indicating lower energy states compared to previous configurations. The highlighted **Zoom #2** region provides a closer inspection of the energy spectrum, enabling precise analysis of closely spaced eigenvalues. Lower and more compact energy levels suggest stronger carrier localization and modified electronic behavior due to the multilayer design. This simulation is

useful for optimizing **quantum dot lasers, photodetectors, LEDs, solar cells, and quantum information devices**, where controlling discrete energy levels improves optical efficiency, carrier confinement, and overall device performance.